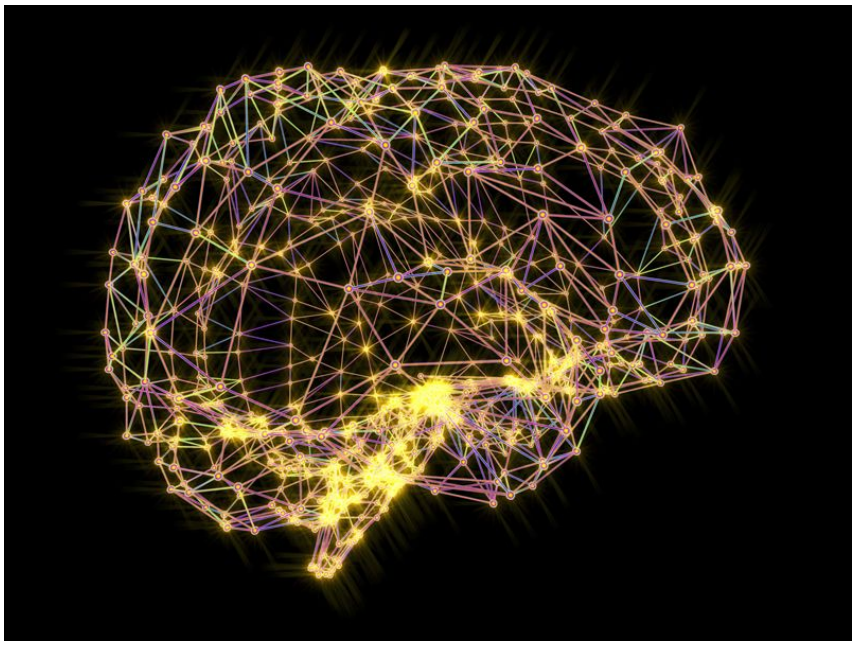

Survival Analysis

Final project

# STDP-based spiking deep convolutional neural networks for object recognition

## Research insights


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Reference :

STDP-based spiking deep convolutional neural networks for object recognition. Neural Networks (2017)

Kheradpisheh, S.R., Ganjtabesh, M., Thorpe, S.J., Masquelier, T

`https://arxiv.org/pdf/1611.01421.pdf`

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As a trial to bridge the gap between deep learning, machine learning, and neuroscience, the referenced paper introduces a Spike Timing Dependent Plasticity (STDP)-based Spiking Deep Neural Network (SDNN). The model uses a more complex structure (multiple layers) than the one used in previous studies, and the goal is to detect visual features or do object recognition. Details about the model and the theory behind it can be found in the paper summary.

In this paper, we will focus on two interesting potential research topics. In fact, Spiking Neural Networks are considered to be the third generation of neural networks. It was shown that they have more computational power per neuron than networks from the previous generations. Even though their accuracy on typical benchmarks such as MNIST, CIFAR, or ImageNet does not reach the same levels as their machine learning counterparts, spiking networks model biology, which means we should expect them to be optimised for the behaviourally most relevant tasks, such as making decisions based on continuous input streams while moving in the real world. Image classification corresponds to the task of classifying a random image suddenly flashed on the retina, without any supporting context. While brains are able to solve such tasks, they are certainly not optimised for it. The main interest for studying spiking neural networks is, however, their close resemblance with biological neural networks. This permits drawing inspiration from experimental neuroscience when designing neural models, and using the knowledge gained from simulations and theoretical analysis of the models to better understand the activity of the brain.

Hence, how can SNNs be optimised time and complexity wise in the context of behaviourally relevant tasks ? Also, why not trying to apply STDP, a powerful and interesting notion, to another type of machine learning in order to better understand the human learning process ?

## Data: The first learning step

SNNs have been mainly tested on image classification data sets, and more generally, on conventional AI benchmarks. There is actually a lack in both good benchmark data sets and evaluation metrics that could measure efficient real-world performance.

To tackle this problem, one potential research direction is the collection of dynamic vision sensor (DVS) benchmarks. A dynamic vision sensor encodes temporal contrast of light intensity into address-events that are asynchronously transmitted for subsequent processing.

The pixels in DVS independently respond to the luminance changes with output spikes. The address encodes the positive or negative brightness changes (ON or OFF) with one bit and the rest of the bits encode the row and column addresses of the triggering pixel. Thus, the spike stream conveys the (x,y) addresses, the firing time, and the polarity (ON/OFF).

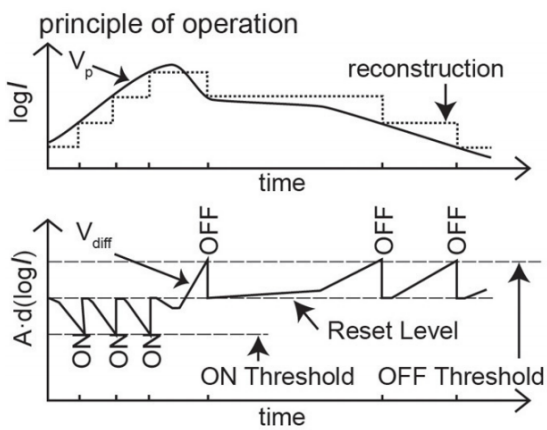


Figure 1: Illustration of how the ON and OFF events are triggered by fixed-size changes of log intensity

DVS is relevant to this topic for two reasons. First, it is a bio-inspired camera since it tries to mimic the behaviour of the biological retina. Second, it is based on the principle of spikes, which is very similar to the way SNNs work. Those reasons make DVS a good fit for real-world behavioural interactions, and that is exactly the area where SNNs would be relevant.

Thus, using DVS for benchmarks collection and then using those benchmarks to train SNNs and evaluate them might give us more insights on human behaviour and learning process. That's why this seems to be an interesting potential research direction.

### STDP-based DSNN and RL: An enhanced approach

As mentioned before, STDP-based DSNNs would be more relevant in behavioural and human oriented tasks since they are based on a biological concept. Using them as a tool to understand the human brain is an interesting research topic. Thus, why not use STDP-based DSNNs in order to understand how the human brain would interact with a new environment ?

When we talk about interaction with a new environment in the context of artificial intelligence, reinforcement learning (RL) is one of the first ideas which come to mind. Reinforcement learning refers to goal-oriented algorithms, which learn how to attain a complex objective (goal) or maximise along a particular dimension over many steps; for example, maximise the points won in a game over many moves. They can start from a blank slate, and under the right conditions they achieve performances which are beyond the human scope. These algorithms are penalised when they make the wrong decisions and rewarded when they make the right ones. This penalty and award process is what is called reinforcement.

Combining deep learning and reinforcement learning has been proven very powerful. The most notorious example is gaming; deep reinforcement learning algorithms can beat human experts playing numerous Atari video games. This powerful combination can be used in human brain inspired algorithms (like SNNs) in order to build new algorithms which are closer to the human brain. Indeed, reinforcement learning adopts a human way of learning (award and penalty method), while SNNs mimic the human brain's way of processing information, and finally, deep learning makes the learning more powerful in a way which can lead to superhuman results.

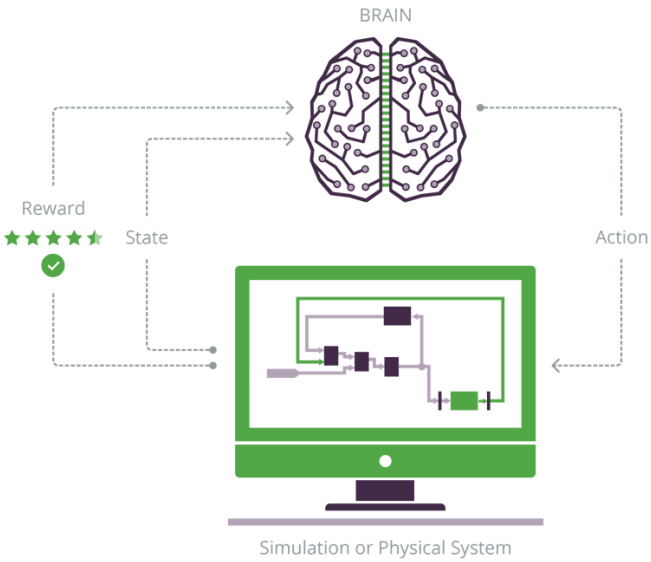


Figure 2: Illustration of deep reinforcement learning

Building algorithms where an agent controlled by a deep spiking neural network learns by interacting with an environment, can give us a better understanding of the human brain learning

process, and allow us to outrank human performances.

Another reason encouraging the combination of those two techniques is the fact that RL is based on agents interacting with the real world. In such situations, temporal information on different time-scales plays an important role, because critical situations require short reaction times hardly accessible by frame-based perception. Deep SNNs have the important property of providing good early estimates, which improve when given more processing time. Although mechanisms to provide early estimations are also proposed for conventional DNNs [1], their implementations are rather artificial and not as seamlessly integrated as in SNNs.

## A simple experiment

In order to evaluate the power of combining deep RL and SNNs, a (very simple) three layer DSNN was built to compute on spikes encoded from Space Invaders observations. The result of this computation is some spiking activity in the out layer, which are converted into actions in the game's action space. The simulation of both the network and the environment are interleaved and appear to operate in parallel, and the network is able to adjust its parameters online using a reward-modulated STDP update rule.

Below (figure 3) are some accompanying plots for a custom DSNN which interacts with the Space Invaders reinforcement learning environment. Spikes (bottom left) and voltages (bottom right) of all neuron populations are plotted.

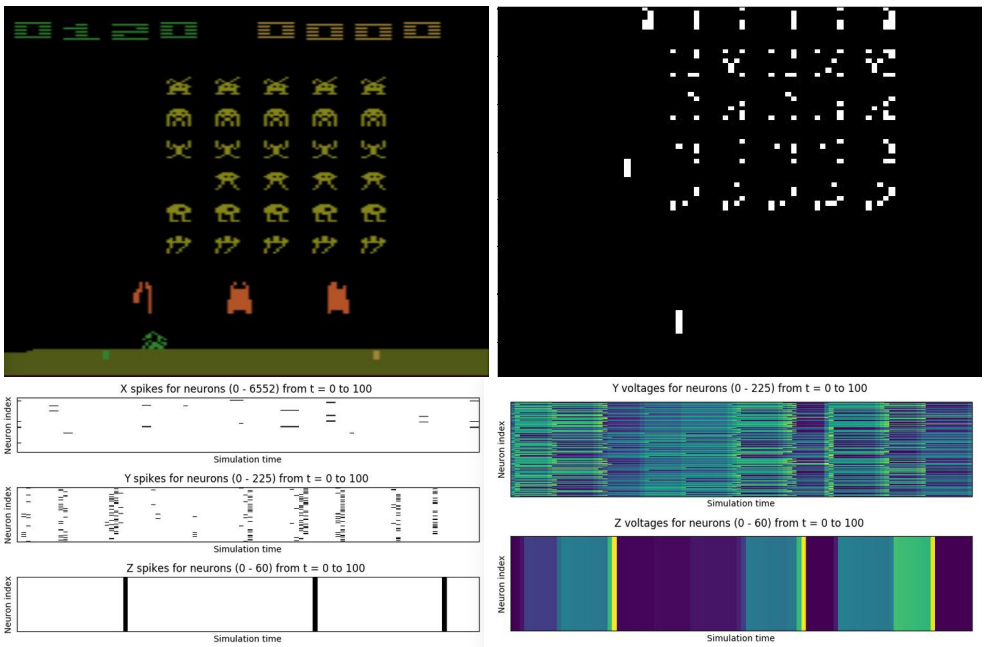


Figure 3: Some plots for RL with DSNN experiment

This simple experiment is not new since it is based on a model which already exists, but it shows that RL and DSNNs can actually be combined, and that the suggested method is not based on unreasonable assumptions. Using this technique in other contexts is, then, very possible and seems to be a direction worth exploring.

[1] Eg., Teerapittayanon et al., 2016

## An interesting combination

In the previous sections, we talked about two points: Data collection using DVS sensors, and the use of RL along with DSNNs. Those two ideas are not independent, and combining them gives a new promising research direction: Using RL and DSNNs on data sets extracted from event-based sensors (like DVS sensors). Indeed, further opportunities for deep SNNs will arise when appropriate benchmark data sets recorded with event-based sensors become available. Large and challenging neuromorphic data sets are not available, yet, but we see a positive trend and increased awareness of the community. Several benchmarks (automated driving, robotics, etc.) have been released, and together with convincing results on problems where conventional systems struggle. This will probably lead to an increasing demand for efficient event-based post-processing systems. Fully event-based systems are not only energy-efficient, but could also better exploit the rich temporal dynamics of the real world than frame-based approaches, which artificially introduce time steps through sensing or processing components.

The combination DVS + RL + STDP DSNNs is powerful and interesting since:

- It is an event based combination
- It has enhanced performances thanks to DL
- It adopts a human brain-like learning process
- It is based on unsupervised learning
- It learns on real-world event based benchmarks

It is like giving a "brain" (represented by the SNN) to a machine which can mimic humans in data collection, learn without supervision, explore new environments while learning from its mistakes (RL), and outrank human performances thanks to the deep learning dimension.

This research direction is very interesting since it can be used in different domains, as well as in contexts where reacting with the real world is needed (neuroscience, self-driving cars, surgeries, robotics, planning, etc.).

## Further insights

There are other potential future directions of research, like the incorporation of recurrence into deep SNNs improving the storage and integration of temporal information. Recurrent SNNs have shown remarkable performance in sequence recognition [2] and generation tasks [3]. Recent work [4] have shown how standard recurrent network architectures such as long short-term memory networks [5] can be ported into the spiking domain, whereas a way to process event-based data with otherwise standard recurrent networks had been presented [6]. Combining recurrent architectures with the intrinsic short-time memory of spiking neurons appears as a promising route for efficiently solving real-world pattern recognition tasks.

[2] Zhang et al., 2015; Panda and Srinivasa, 2018
[3] Rajan et al., 2016; Panda and Roy, 2017
[4] E.g., Diehl et al., 2016; Bellec et al., 2018
[5] LSTMs, Hochreiter and Schmidhuber, 1997
[6] Neil et al., 2016